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- Problem/Objective
 - End-to-End Autonomous Driving
- Contribution/Key Idea
 - Multi-trajectory + World model
 - Real-time (latency: 17~18ms)
 - open-loop/closed-loop에서 둘다 SOTA

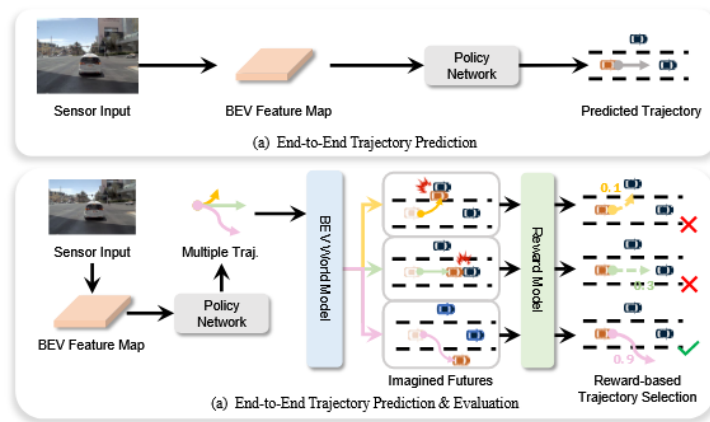
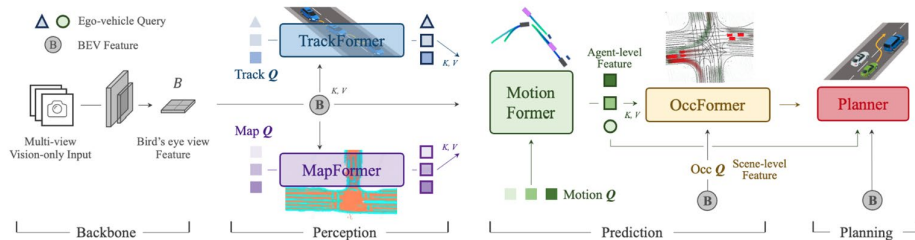


Figure 1. **The illustration of end-to-end trajectory evaluation with BEV world model.** (a) Previous end-to-end driving methods primarily focus on learning a high-quality trajectory. (b) Our WoTE consists of end-to-end trajectory prediction and evaluation. First, we predict multi-modal trajectories. Next, given these trajectories, we leverage the BEV world model to predict the corresponding future BEV states. The reward model evaluates these trajectories based on the predicted states and assigns rewards, selecting the highest-reward trajectory.

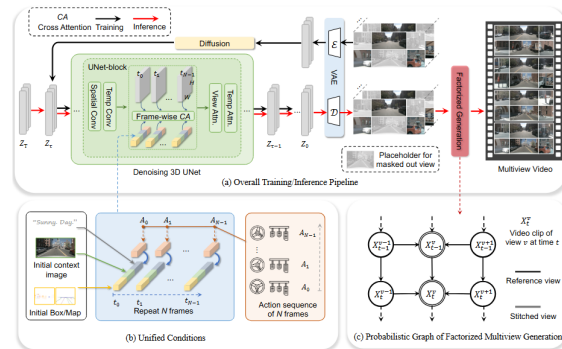
● Introduction

[미래 장면을 예측하지 않음 (only t시점)]



UniAD [1]

[multi-future supervision 부족]



Driving into the Future [2]

● 기존 방법들의 한계점 분석 [1], [2] etc.

- 기존은 perception 결과 or 현재 state에만 주목 → 각 후보 trajectory가 미래에 어떤 결과인지 학습 x → 오류 누적
- 미래 예측 x : only t 시점 supervision → **multi-future supervision 부족**
- 이를 cover 하기 위해 t+c 시점 생성 모델들 존재

■ Mostly diffusion → 실시간 불가 → **lightweight 한 world model 로 학습하자!**

[1] Hu, Yihan, et al. "Planning-oriented autonomous driving." *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2023.

[2] Wang, Yuqi, et al. "Driving into the future: Multiview visual forecasting and planning with world model for autonomous driving." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024.

- Introduction

- 본 연구의 아이디어

- 1) **BEV 꿈 / 엇샷 world model** 끝쵸

- a) image-level보다 훨씬 compact

- b) single-step transformer encoder만으로 미래 BEV state 예측(실시간성 good)

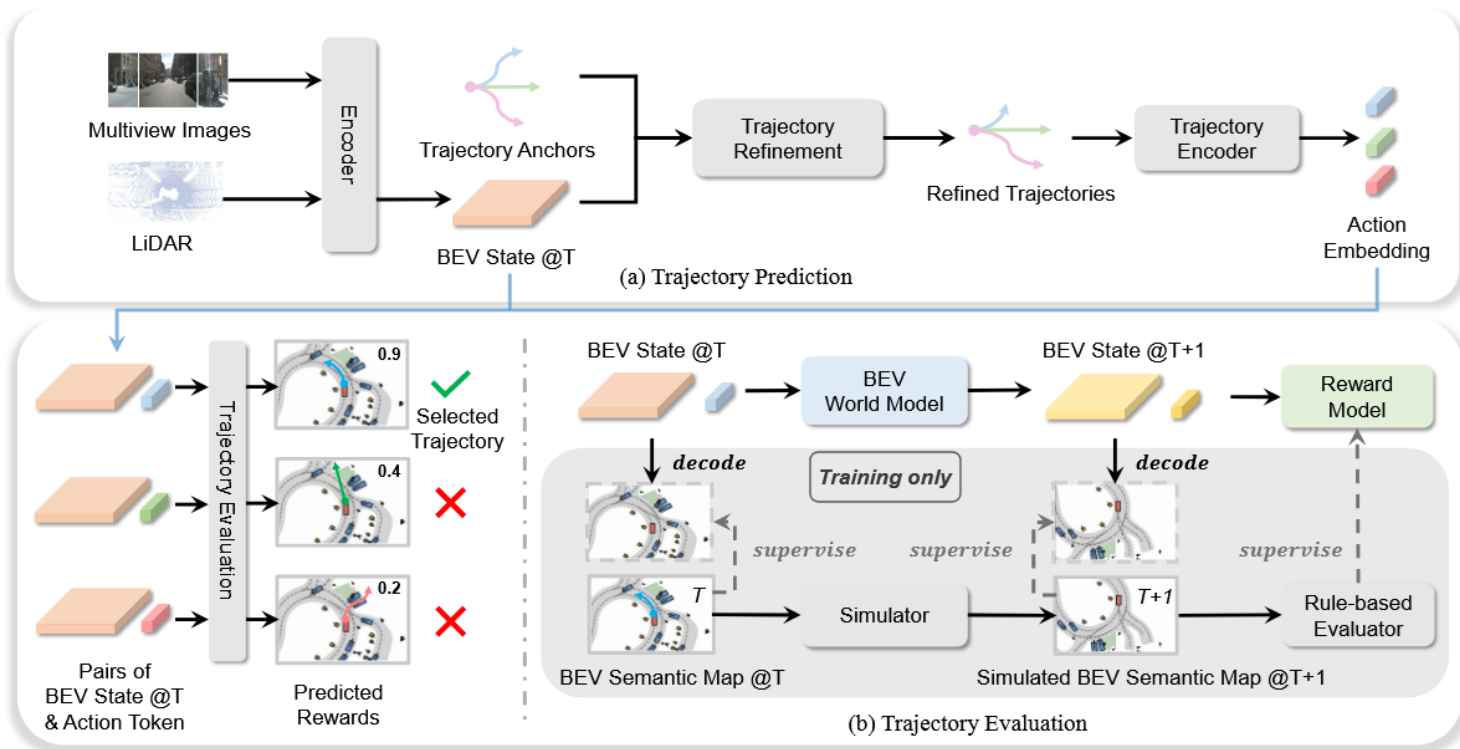
- 2) **Off-the-Shelf BEV simulator** ~~렌~~ supervision

- a) multi-trajectory여도 supervision 가능

- 3) **BEV states** ~~긱~~ **reward model**

- a) imitation reward + simulation reward로 구성 trajectory quality를 계산

Framework

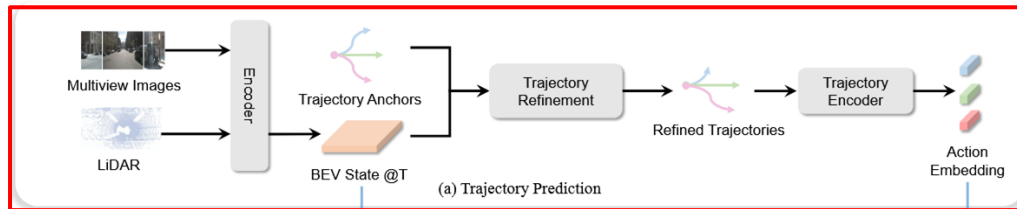


Input : L + C / Output : (Best) Trajectory

● Method

○ Trajectory Predictor

- Input : L+ C
- Output : Action embedding



- BEV encoder : TransFuser [3] 사용
- Trajectory anchor
: 기존 방식
 - 학습 데이터셋의 expert trajectory → K-Means clustering
- Trajectory refinement
: anchor trajectory들에서 BEV 정보를 고려하여 refinement
 - τ : trajectory, **TE**: trajectory encoder, **B**: BEV state (feature map)

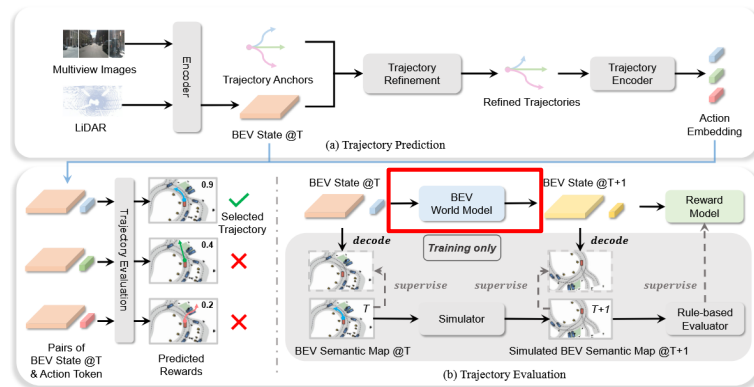
$$\hat{\tau} = \tau + \text{MLP}(\text{CrossAttention}(\text{TE}(\tau), \mathbf{B}, \mathbf{B})),$$

Method

BEV world model

- Input : BEV state + action embedding
- Output : (refined) BEV state + action embedding

$$(\mathbf{B}_t, \mathbf{a}_t^i) \quad \mathbf{A}_t = \{\mathbf{a}_t^1, \mathbf{a}_t^2, \dots, \mathbf{a}_t^N\}$$



BEV World Model: (t+1 시점의) 8x8 BEV state를 예측하는 모델

- Input: state-action pair = BEV state + action embedding(encoding된 trajectory) 사용

$$(\mathbf{B}_{t+1}^i, \mathbf{a}_{t+1}^i) = \text{WorldModel}(\mathbf{B}_t, \mathbf{a}_t^i).$$

Recurrent Future Prediction

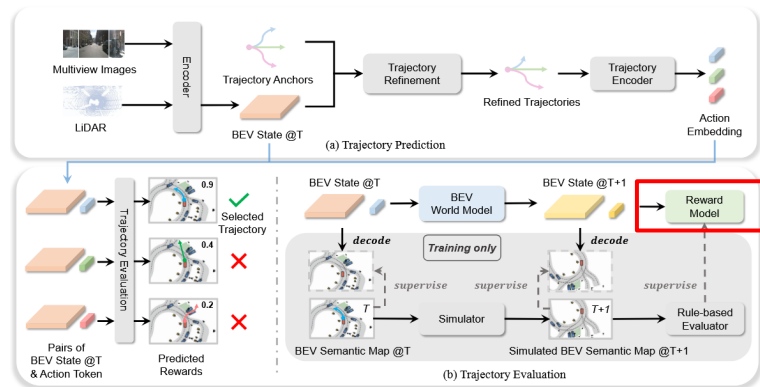
- Future K step을 recurrent manner로 예측

$$(\mathbf{B}_{t+K}^i, \mathbf{a}_{t+K}^i) = \text{WorldModel}(\mathbf{B}_{t+K-1}^i, \mathbf{a}_{t+K-1}^i).$$

Method

Reward model

- trajectory 후보($\hat{r}_1 \dots \hat{r}_N$)에 대해
“현재 + 미래 BEV state sequence”를 기반으로
각 trajectory의 reward score를 예측하는 모델
- Input : 미래 BEV state sequence / action embedd
Output : trajectory의 reward vector



$$\mathbf{r}_i = \text{RewardModel}(\mathbf{B}_{\text{all}}^i, \mathbf{a}_{\text{all}}^i),$$

$$\begin{aligned} \mathbf{B}_{\text{all}}^i &= \text{cat}([\mathbf{B}_t, \mathbf{B}_{t+1}^i, \dots, \mathbf{B}_{t+K}^i]) \\ \mathbf{a}_{\text{all}}^i &= \text{MLP}(\text{cat}([\mathbf{a}_t^i, \mathbf{a}_{t+1}^i, \dots, \mathbf{a}_{t+K}^i])) \end{aligned}$$

- Reward: Imitation + simulation

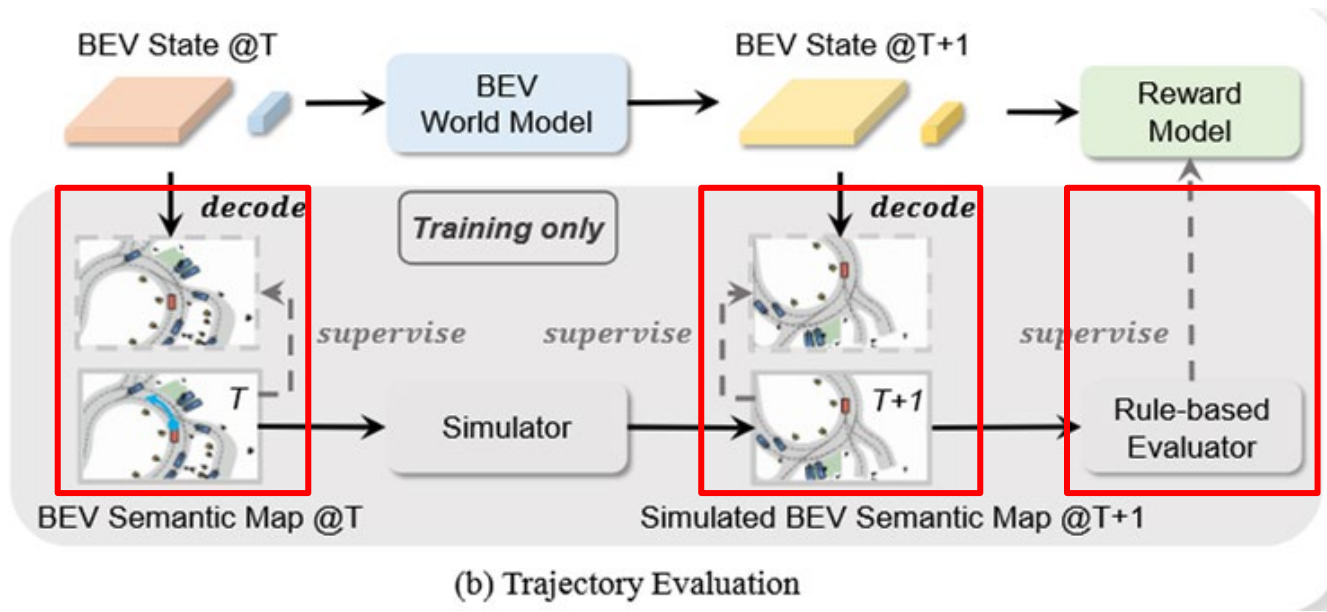
현재와 미래 state를 고려하여 reward prediction

$$\begin{aligned} r_{\text{final}} = - \bigg(&w_1 \log r_{\text{im}} + w_2 \log r_{\text{sim}}^{\text{NC}} + w_3 \log r_{\text{sim}}^{\text{DAC}} \\ &+ w_4 \log (5r_{\text{sim}}^{\text{TTC}} + 2r_{\text{sim}}^{\text{Comf}} + 5r_{\text{sim}}^{\text{EP}}) \bigg), \end{aligned}$$

• Method

○ Multi-Future Supervision

- Traffic simulator인 nuPlan 사용하여 BEV space supervision



$$\mathcal{L}_{\text{BEV}} = \text{FocalLoss}(\mathbb{B}_{t+k}, \mathbb{B}_{t+k}^*),$$

$$\mathcal{L}_{\text{reward}}^{\text{sim}} = \text{BCE}(r_{\text{sim}}, r_{\text{sim}}^*).$$

• Experiment

Method	Input	#Traj.	Traj. Eval.	NC $\uparrow$	DAC $\uparrow$	EP $\uparrow$	TTC $\uparrow$	Comf. $\uparrow$	PDMS $\uparrow$
Human	-	-	-	100	100	87.5	100	99.9	94.8
Constant Velocity	-	1	$\times$	69.9	58.8	49.3	49.3	100.0	21.6
Ego Status MLP	-	1	$\times$	93.0	77.3	62.8	83.6	100.0	65.6
VADv2 [49]	C	8192	Rule-based	97.9	91.7	77.6	92.9	100.0	83.0
UniAD [23]	C	1	Rule-based	97.8	91.9	78.8	92.9	100.0	83.4
LTF [37]	C	1	$\times$	97.4	92.8	79.0	92.4	100.0	83.8
PARA-Drive [49]	C	1	Rule-based	97.9	92.4	79.3	93.0	99.8	84.0
TransFuser [37]	C & L	1	$\times$	97.7	92.8	79.2	92.8	100.0	84.0
LAW [29]	C	1	$\times$	96.4	95.4	81.7	88.7	99.9	84.6
DRAMA [52]	C & L	1	$\times$	98.0	93.1	80.1	94.8	100.0	85.5
Hydra-MDP [31]	C & L	8192	Model-free	98.3	96.0	78.7	94.6	100.0	86.5
WoTE	C & L	256	Model-based	98.5	96.8	81.9	94.9	99.9	88.3

Table 1. **Comparison with the SOTA approaches on NAVSIM test set.** Rule-based: UniAD and PARA-Drive use occupancy maps, while VADv2 relies on vectorized maps for trajectory evaluation following specific rules. Model-free: Hydra-MDP evaluates trajectory without world modeling. Model-based: our WoTE evaluates trajectory with the assistance of the BEV world model. NC: no at-fault collision. DAC: drivable area compliance. EP: ego progress. TTC: time-to-collision. Comf.: comfort. PDMS: the predictive driver model score. Traj.: Trajectory. Eval.: Evaluation. C: Camera. L: LiDAR.

- Experiment

Method	Traj. Eval.	Success Rate↑	Driving Score↑
AD-MLP [53]	-	0.00	18.05
UniAD [23]	Rule-based	16.36	45.81
VAD [26]	Rule-based	15.00	42.35
TCP [50]	-	30.00	59.90
WoTE	Model-based	31.36	61.71

Table 2. **Comparison with SOTA approaches on closed-loop Bench2Drive [25] benchmark on CARLA simulator.** Traj.: Trajectory. Eval.: Evaluation.

Traj. Eval.	Future States	NC ↑	DAC ↑	EP ↑	TTC ↑	Comf. ↑	PDMS ↑
×	×	96.4	91.5	76.2	90.3	99.9	81.0
✓	×	97.1	92.9	78.2	91.9	100.0	83.2
✓	✓	98.0	94.7	79.9	93.4	100.0	85.6

Table 3. **Ablation study on trajectory evaluation and future state prediction.** Traj: Trajectory. Eval.: Evaluation.

- Experiment

Future Prediction Steps	NC $\uparrow$	DAC $\uparrow$	EP $\uparrow$	TTC $\uparrow$	Comf. $\uparrow$	PDMS $\uparrow$
0s $\rightarrow$ 4s	97.4	93.6	79.1	91.9	100.0	84.0
0s $\rightarrow$ 2s $\rightarrow$ 4s	98.0	94.7	79.9	93.4	100.0	85.6

Table 5. **Ablation study on the different number of time steps in the future prediction.** Our BEV world model predicts future states in a recurrent manner. A $\rightarrow$ B: the predicted future state at time B is based on the state at time A.

#Trajs.	NC $\uparrow$	DAC $\uparrow$	EP $\uparrow$	TTC $\uparrow$	Comf. $\uparrow$	PDMS $\uparrow$	Latency $\downarrow$
64	97.4	94.1	79.6	91.9	100.0	84.5	17.2 ms
128	97.9	94.6	80.0	93.0	100.0	85.4	17.9 ms
256	98.0	94.7	79.9	93.4	100.0	85.6	18.7 ms

Table 6. **Ablation study on the different number of trajectories.** The number of trajectories is the same during training and testing. The latency is evaluated on an NVIDIA L20 GPU. Trajs.: Trajectories.

- Experiment

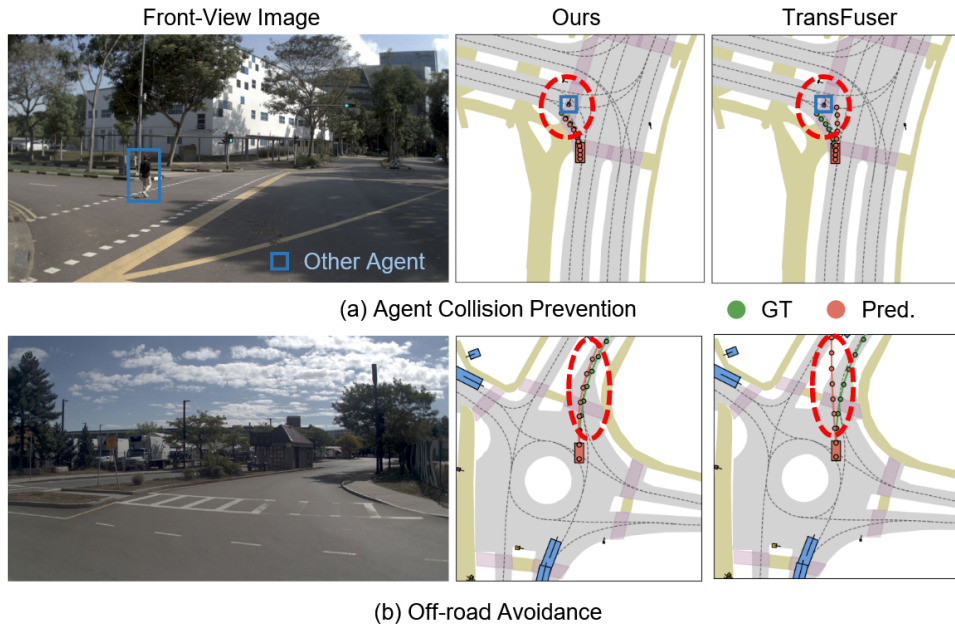


Figure 3. **Visualization of end-to-end planning trajectories.** We compare our method with TransFuser [37], which serves as our trajectory prediction baseline without a trajectory evaluation module. Our trajectory evaluation module effectively filters out low-quality trajectories and helps avoid collisions.